

Enhanced Perception and Performance by Multimodal Threat Cueing in Simulated Combat Vehicle

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Objective: In a simulated combat vehicle, uni-, bi-, and trimodal cueing of direction to threat were compared with the purpose to investigate whether multisensory redundant information may enhance dynamic perception and performance.

Background: Previous research has shown that multimodal display presentation can enhance perception of information and task performance.

Method: Two experiments in a simulated combat vehicle were performed under the instructions to turn the vehicle toward the threat as fast and accurately as possible after threat cue onset. In Experiment 1, direction to threat was presented by four display types: visual head-down display, tactile belt, 3-D audio, and trimodal with the three displays combined. In Experiment 2, direction to threat was presented by three display types: visual head-up display (HUD)–3-D audio, tactile belt–3-D audio, and trimodal with HUD, tactile belt, and 3-D audio combined.

Results: In Experiment 1, the trimodal display provided overall best performance and perception of threat direction. In Experiment 2, both the trimodal and HUD–3-D audio displays led to overall best performance, and the trimodal display provided overall the best perception of threat direction. None of the trimodal displays induced higher mental workload or secondary task interference.

Conclusion: The trimodal displays provided overall enhanced perception and performance in the dynamically framed threat scenario and did not entail higher mental workload or decreased spare capacity.

Application: Trimodal displays with redundant information may contribute to safer and more reliable peak performance in time-critical dynamic tasks and especially in more extreme and stressful situations with high perceptual or mental workload.

Keywords: trimodal display, bimodal display, multimodal, threat cueing, tactile display, 3-D audio, visual display, perception and performance

INTRODUCTION

Fundamental to multisensory information is the information redundancy we frequently and unconsciously process and capitalize on in our natural perception and action (e.g., Ernst & Bühlhoff, 2004). Multisensory information that is complementary or redundant can improve detection and alerting effects (e.g., Spence & Driver, 2004), and multisensory displays may therefore support and improve operator perception and performance (Ho & Spence, 2008). There is in essence a need for further empirical studies on multisensory or multimodal processing in ecologically relevant situations, which include the interplay between endogenous (top-down) and exogenous (bottom-up) attention control, to realize the full potential of multimodal display interface design (e.g., Sarter, 2006, 2007; Spence & Ho, 2008). Although further empirical studies are needed, guiding principles for multimodal information in dynamic operator tasks are provided by multiple resource theory (Wickens, 1984, 2008; Wickens & Hollands, 2000). It suggests that we have several independent resources for processing of information but that some resources are more suited for simultaneous use (parallel processing) than others (sequential processing). That is, tasks using compatible resources that allow parallel processing may usually be performed simultaneously. Intramodal competition of resources within one modality may however cause interference, which may significantly deteriorate performance if one modality handles more than one task (e.g., simultaneous handling of two auditory tasks).

Santangelo and Spence (2007) performed experiments with unimodal auditory and visual cueing, respectively, and bimodal audiovisual cueing of peripheral visual targets. Better

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HUMAN FACTORS

Vol. 54, No. 1, February 2012, pp. 122-137

DOI:10.1177/0018720811424895

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performance was demonstrated with bimodal cueing during high perceptual load, whereas no difference was found during low perceptual load. This indicates the key role of multisensory interaction in disengaging spatial attention from perceptually demanding tasks and suggests a more efficient capture of attention by multisensory warning signals in conditions of high perceptual load (Santangelo & Spence, 2007). As for including auditory cueing in multimodal displays, 3-D audio technology makes it possible to present sounds in 3-D, which can increase the ability to detect and interpret one specific sound source among others (e.g., Ericson & McKinley, 1997) and is useful in noisy environments because of the spatial separation of sound (Begault, 1999; Nelson, Bolia, Ericson, & McKinley, 1998). One drawback with 3-D audio, however, is front-back confusion, which simply is the misconception of a sound coming from straight ahead as coming from behind or vice versa (e.g., Weightman & Kistler, 1999). Still, 3-D audio can be beneficial, and, for example, combined 3-D audio and visual displays have improved target acquisition in fighter aircraft (Veltman, Oving, & Bronkhorst, 2004).

Besides the more obvious visual and auditory displays, tactile displays have demonstrated positive effects on human perception and performance in various environments and tasks. For example, tactile displays have proven beneficial for conveying spatial orientation information (McGrath, Estrada, Braithwaite, Raj, & Rupert, 2004; van Erp, Groen, Bos, & van Veen, 2006), including information for helicopter hovering (van Erp, Veltman, & van Veen, 2003) and helicopter landing on a ship's deck (Jennings, Craig, Cheung, Rupert, & Schultz, 2004) and have also effectively supported waypoint land navigation (Eriksson et al., 2008; Redden, 2006; van Erp, van Veen, Jansen, & Dobbins, 2005). The inclusion of tactile displays has been shown to be particularly helpful in situations with high cognitive demands, visual channel overload, or redundant cueing (e.g., Gilson, Redden, & Elliott, 2007; Redden, 2006; Spence & Ho, 2008; van Erp, 2007). For example, redundant threat cueing with combined visual and tactile displays has improved response time and intercept time compared to

only visual cueing in simulated aerial combat with fighter aircraft including high acceleration (van Erp et al., 2007).

In Carlander and Eriksson (2006) 3-D audio and tactile threat cueing were explored in the combat vehicle CV90, which is equipped with sensors for various functions such as missile warning, laser warning, and optical surveillance. The sensor information is brought together in a warning and countermeasures system with threat warnings presented on a head-down display (HDD) combined with an alerting sound. The alerting sound does not provide information of threat direction, and the driver of the combat vehicle must look down on the HDD to perceive the direction to the threat. This entails that crucial time may be lost in building up the spatial awareness necessary for avoidance maneuvering, defense positioning, and other relevant tasks. Although a 3-D audio display was used by Carlander and Eriksson (2006) as remedy for the nondirectional auditory alert, several occurrences of front-back confusion appeared with unimodal presentation of 3-D audio. However, the combination of 3-D audio and tactile displays neutralized the front-back confusions and improved the prompt perception of direction to threat in the CV90. The scenario in Carlander and Eriksson included the driver tasks of responding as quickly as possible to threat cue onsets by maneuvering toward threat and aligning vehicle heading as exactly as possible with threat. The different phases of the used threat scenario put different requirements on the operator depending on whether it was the initial phase of quickly gaining spatial awareness of direction to threat, the subsequent maneuvering toward threat, or the final phase of exactly aligning vehicle with threat. Although the tactile and 3-D audio displays improved the immediate perception of threat direction and effectively provided information for vehicle maneuvering, such displays are most likely outperformed in precision tasks by visual displays that properly capitalize on the superior spatial resolution of the visual sense.

As follow-up to Carlander and Eriksson (2006), the present study includes the previously used threat scenario in two experiments with threat cueing made in a simulated combat

vehicle and delivered by displays employing vision, audition, and touch. A pragmatically oriented approach on the combination of endogenous and exogenous attention control is applied to multimodal or multisensory threat cueing in the simulated combat vehicle. Primarily four aspects reflect this pragmatically oriented approach. First, emphasizing the endogenous attention control mode, the participants were briefly primed for expecting threat warning in the employed display interfaces and sensory modalities by means of short training or familiarization sessions prior to the experiments. Second, emphasizing the exogenous attention control mode, the display interfaces and sensory modalities were combined to support redundancy in which the sensory processing concerns the same or very similar information of the direction to threat or target. Third, a secondary task was introduced with the purpose to increase perceptual or mental workload, to some extent reflect the potentially demanding task environment in a combat vehicle, and to provide a secondary measure of mental workload. Fourth, for each of the employed displays only one display interface design was investigated: one design each for HDD, head-up display (HUD), tactile belt, and 3-D audio. Experiment 1 investigated unimodal cueing of threat direction with HDD, 3-D audio, and tactile belt and trimodal cueing of threat direction with the combination of these three displays. Experiment 2 investigated bimodal cueing of threat direction with HUD combined with 3-D audio and tactile belt combined with 3-D audio and trimodal cueing of threat direction with the visual, tactile, and auditory displays combined.

EXPERIMENT 1

The combining of the displays into a trimodal display configuration may show multisensory benefits that go beyond the benefits of unimodal and bimodal displays. We hypothesized that there might be further benefits of the trimodal display because of the displays' different potential advantages in different phases of threat direction cueing and maneuvering relative to a threat. In addition, there might be further benefits of the trimodal display in terms of alerting an operator and

speeding up operator response, especially during heightened perceptual and/or cognitive load. Along these lines, in the applied setting of a simulated combat vehicle, the overall purpose of Experiment 1 was to investigate cueing of threat direction by a trimodal display consisting of HDD, 3-D audio and tactile belt as compared with each of the unimodal counterparts. Although threat direction cueing with both the HDD and the tactile belt was unimodal, the displays were both paired with a nondirectional auditory alert. Thus, the conditions including the HDD and the tactile belt were both bimodal in terms of warning information but unimodal in terms of threat direction information. The rationale was that warning of prioritized threat in the combat vehicle will not exclude auditory alert.

Based on the displays' different potential advantages in different phases of the threat cueing scenario, the main hypotheses were that the trimodal cueing of threat direction would lead to the overall best performance and that it would be perceived as overall the best for perception of threat direction. One concern, however, was whether the trimodal display might add mental workload, or interference, compared with unimodal directional cueing, and especially so with a secondary task introduced.

Furthermore, Wickens and Gosney (2003) specify five different redundancy effects of combined visual and auditory display information that can be used in assessing effects of multimodal displays. The effects are (a) *gestalt*—the whole is better than the parts (performance with multimodal display is better than each included modality), (b) *best of both worlds*—performance with multimodal display is equal to single included modality with best performance, (c) *averaging*—performance with multimodal display is equal to the mean performance of included modalities, (d) *worst of both worlds*—performance with multimodal display is equal to single included modality with the worst performance, and (e) *antigestalt*—performance with multimodal display is worse than performance with each included modality. Hence, these redundancy effects were used as a conceptual framework in the assessment of the effects of multimodal displays.

Method

Participants. In all, 12 naïve participants, all males with normal sight and hearing, participated. The mean age was 23.0 years (21–29 years).

Apparatus and stimuli. A tactile torso belt developed at the Swedish Defence Research Agency was used for tactile cueing. The belt was worn on the torso and contained 12 DC motor tactors placed according to clock positions in which each tactor covered 30° of the horizontal dimension at 0°, 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, and 330°. The 3-D audio threat cueing, mono sound, and radio calls (see below) were presented through Senheisser HD 200 studio headphones with a semiopen circumaural design and frequencies of 15 to 22,000 Hz. The 3-D sound was generated by a Hercules Muse pocket soundcard running the Sensaura algorithm with general head-related transfer functions (HRTFs). An Ascension LaserBIRD II headtracker with six degrees of freedom, 0.7 mm RMS accuracy, and sampling frequency of 240 Hz was used to compensate for head movements in the 3-D audio presentation. The visual threat cueing was presented on a 15-in. HDD LCD display with a spatial resolution of 1,024 × 780 pixels placed at waist height approximately 30° left of the center.

The out-of-the-window view was presented on a 37-in. plasma screen with spatial resolution of 1,280 × 960 pixels (see Experiment 2, Figure 5). Response of radio calls was given on a 17-in. touch LCD screen with spatial resolution of 1,280 × 1,024 pixels placed at torso height approximately 45° right of the center. A plastic frame was mounted in front of the touch screen, giving it a visible area of 21.5 × 16.0 cm. The touch screen presented six buttons in two columns named “external” and “internal.” Each button had a specific color, with the name of the color printed on the button.

The simulation was performed on a Moog 6DOF200E motion platform and supported by a simulation engine, developed at the Swedish Defence Research Agency, for the visual and motion cueing of vehicle dynamics in the simulated terrain.

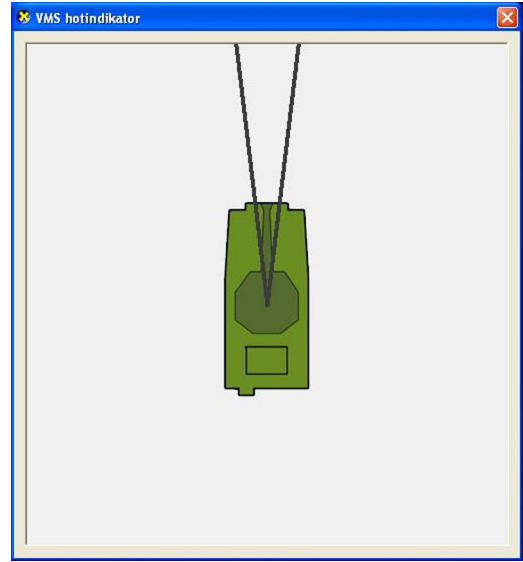


Figure 1. Visual head-down display (HDD), with threat indicator pointing toward a threat in front of the vehicle.

The experiment had a 4 (display) × 3 (sector) within-subjects design. The four display conditions for cueing of threat direction were HDD, tactile belt, 3-D audio in headphones, and these three displays combined into a trimodal display. For the HDD and tactile displays, the sound alert was given in mono (i.e., with no directional information; see Table 1).

Threat direction was visually indicated by a 15°-wide wedge-shaped threat indicator on the HDD as illustrated in Figure 1. The front of the threat indicator always pointed in the direction of the front of the vehicle. Thus, with threats being stationary, the wedge in the threat indicator dynamically rotated in accordance with vehicle rotation.

The tactors of the tactile belt were individually activated one at a time with an approximate frequency of 120 Hz, and followed a repeated pattern of 90–70 ms on–off; 100–20 ms on–off; 100–120 ms on–off. The on–off pattern was introduced to minimize the risk of adaptation to a continuous on signal, and a minimum of 90 ms on signal was used to make sure that the tactor reached its maximum frequency before turning it off. The sound for audio warning and threat cueing was similar to

TABLE 1: The Four Types of Display Conditions With Type of Threat Cueing, Sound Alert, and Radio Calls

Display	Threat Direction Cueing	Sound Alert	Radio Calls
HDD	HDD	Mono sound	Mono sound
Tactile	Tactile belt	Mono sound	Mono sound
3-D audio	3-D audio	3-D audio	3-D audio
Trimodal	HDD + tactile belt + 3-D audio	3-D audio	3-D audio

Note. HDD = head-down display.

the beeping sound alert given for laser warnings in the real CV90, a pulsating sound with 446 ms on followed by 461 ms silence, with the prominent frequency of 438 Hz. Both the tactile and auditory patterns were previously used in Carlander and Eriksson (2006).

The three sectors of threat cueing onsets were relative to current vehicle heading, front $0^{\circ} \pm 45^{\circ}$, back $180^{\circ} \pm 45^{\circ}$, and sides $90^{\circ} \pm 45^{\circ}$ and $270^{\circ} \pm 45^{\circ}$ collapsed into one side sector condition (i.e., right and left not treated as experimental conditions). Each participant was exposed to four blocks of trials, one block for each display condition. In each block there were 24 threat cueing onsets with eight threats in each of the three sectors and with positions randomly distributed within each sector. The block order was different for each participant.

The main task was to turn the vehicle as quickly as possible toward the perceived threat direction at threat cueing onset and to push a trigger button when the vehicle heading was perceived to be aligned with the threat. The threats were not presented in the visual surroundings outside the vehicle, but a text message overlaid on the out-of-the-window view saying “hit” or “miss” was given as feedback after each trigger-button response. A “hit” was defined as vehicle heading aligned within $\pm 10^{\circ}$ of the threat position at trigger-button response. Each threat presentation was limited to the duration of 20 s or terminated by the trigger-button response. Threat cueing offset-to-onset intervals were randomized between 15 and 20 s from trigger-button response. Localization error (LE) of response was defined as the absolute value of deviation in degrees of vehicle heading from threat position.

Response time (RT) was defined as elapsed time in ms from threat cueing onset to having turned the vehicle by 10° , or elapsed time to trigger-button response if the threat was presented within $\pm 10^{\circ}$ of the vehicle center.

After the trigger-button response was completed, a secondary task response was performed, which was to identify information in two simultaneously presented radio calls. One radio call began with the call “internal” and the other with “external,” each followed by the name of a color. The secondary task response was to push the matching colored buttons on the touch screen. In this way, auditory, visual, tactile, and motor systems were engaged in the secondary task. The radio calls were presented simultaneously with threat cueing onset. In the HDD and tactile display conditions the radio calls were presented in mono, whereas in the 3-D audio and multimodal display conditions the radio calls were presented in 3-D audio with voices positioned horizontally at -22° and $+22^{\circ}$ from the center (see Table 1). The sound levels and frequencies of the sound alert, 3-D audio threat cueing, and radio calls were tested and established to ensure that no masking of information occurred. The proportion of correct responses to the radio calls was measured and defined as proportion correct radio calls (PCR).

Questionnaires with 7-point rating scales were used to assess the participants’ subjective ratings of mental workload, effort of driving, and quality of perceived threat direction during the different phases of the task (1 = *very low*, 7 = *very high*).

Procedure. The participant was instructed to drive the vehicle along a road in the simulated

terrain. When a threat was indicated, the main task was to turn the vehicle as quickly as possible toward the perceived threat direction and to push the trigger button when the vehicle heading was perceived to be aligned with the threat. The participant was instructed to prioritize and first perform the main task and then to perform the response to the radio calls of the secondary task. After completion of the secondary task, the driving along the road continued until the next threat cueing onset. A short training or familiarization session was performed with two trials of main and secondary tasks for each display condition, after which the four blocks of trials of the experiment proper were performed. The questionnaire concerning mental workload and effort of driving was answered after each block of trials. At the end of the session, the concluding questionnaire concerning perception of threat direction was answered. Total time for each experimental session was approximately 95 min.

Results

In the case of violation of the sphericity assumption in the analyses of variance (ANOVAs), the Greenhouse-Geisser-corrected p value is given (indicated by ^G). All post hoc tests were performed with Tukey's honestly significant difference (HSD) test.

Performance data. Performance data were analyzed with ANOVA for a 4×3 repeated measures factorial design (4 displays $\times$ 3 sectors). For the 3-D audio display presentation in the back sector, three participants experienced occurrences of front-back confusion that resulted in LEs of approximately 180° (on one, two, and three occasions, respectively). These data were treated as outliers and not included in the calculation of the condition mean for these three participants (thus, the condition means for these participants were calculated from 7, 6, and 5 presentations, instead of 8, for the 3-D audio back sector). Before analysis with ANOVA, the PCR data were, because of proportions greater than .80, transformed with the $2\arcsin(x)$ formula to remedy violations against the assumption of normal distribution (Kirk, 1995).

Localization error. The ANOVA showed a significant main effect of display, $F(3, 33) =$

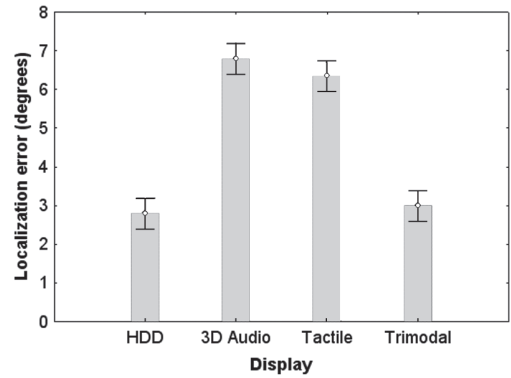


Figure 2. Mean localization error with each display configuration. Error bars = ± 1 standard error; HDD = head-down display.

61.23, $p < .001$, with no other significant effects (Figure 2). Post hoc testing showed a larger mean LE for both the 3-D audio and tactile displays compared with both the HDD and the trimodal display ($ps < .001$). The 3-D audio presentation in the back sector led to front-back confusion in 6 of the 96 (6.3%) presentations (8 presentations $\times$ 12 participants; see above).

Response time. The ANOVA showed a significant main effect of display, $F(3, 33) = 7.93$, $p < .001$, with no other significant effects (Figure 3). Post hoc testing showed a longer mean RT for both the HDD and the 3-D audio compared with both tactile and trimodal displays ($ps < .025$).

Proportion correct radio calls. The ANOVA on $2\arcsin(x)$ -transformed values showed no significant effects of PCR. The overall mean PCR was 0.83 (untransformed).

Subjective ratings. Perception of threat direction was analyzed with ANOVA for a 4×3 repeated measures factorial design (4 displays $\times$ 3 threat phases). The threat phases were at threat cueing onset with first response (initial), during subsequent turning of the vehicle toward threat (middle), and aligning of the vehicle with threat and pushing the trigger (final). Mental workload and effort of driving were analyzed with ANOVA for a one-way repeated measures design (four displays).

Perception of threat direction (PTD). The ANOVA showed a significant main effect of display, $F(3, 33) = 23.46$, $p < .001$, and a

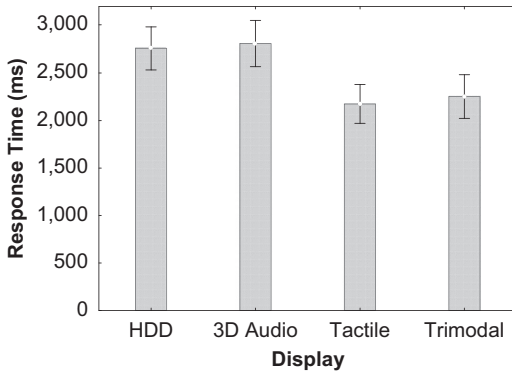


Figure 3. Mean threat response time with each display configuration. Error bars = ± 1 standard error; HDD = head-down display.

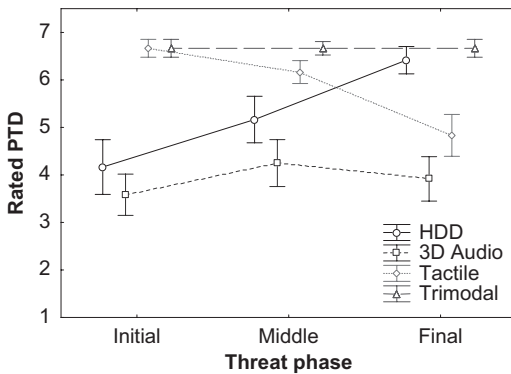


Figure 4. Perception of threat direction (PTD) for the four display types at threat cue onset (initial), maneuvering toward threat (middle), and aligning vehicle with threat (final). Error bars = ± 1 standard error; HDD = head-down display.

significant interaction effect of display by threat phase, $F(6, 66) = 6.58$, $p < .01^G$. There was no significant main effect of threat phase (Figure 4). Post hoc testing showed the following significant differences of the ratings of the display configurations within each threat phase: initial—both trimodal and tactile higher rated than both HDD and 3-D audio ($ps < .001$); middle—both trimodal and tactile higher rated than 3-D audio ($ps < .01$); and final—trimodal higher rated than both tactile ($p < .025$) and 3-D audio ($p < .001$) and HDD higher rated than 3-D audio ($p < .01$). Post hoc testing also showed that the

HDD was significantly lower rated in the initial phase compared with its use in the final phase ($p < .001$) and, in contrast, that the tactile display was significantly higher rated in the initial phase compared with its use in the final phase ($p < .025$). Post hoc testing of the main effect of display showed that the trimodal was rated higher than both the HDD and 3-D audio displays ($ps < .01$) and that the 3-D audio was rated lower than all other displays ($ps < .01$).

Mental workload. The ANOVA showed no main effect of display. The ratings of mental workload were moderate for all four displays with an overall mean of 4.0 on the 7-point rating scale (1 = very low, 7 = very high).

Effort of driving. The ANOVA showed no main effect of display. The ratings of effort of driving were low for all four displays with an overall mean of 2.3 on the 7-point rating scale (1 = very low, 7 = very high).

Conclusions

First, the main hypotheses were supported. Trimodal threat cueing provided overall best performance, explained by significantly better precision compared with the tactile and 3-D audio displays, respectively, and shorter RT compared with the HDD and the 3-D audio, respectively. In terms of redundancy effects, the trimodal display is the best of both worlds because of the nonsignificant difference with the HDD for LE and tactile belt for RT. The trimodal display also provided the overall best PTD, by higher ratings compared with the HDD and the 3-D audio, respectively, in the initial threat phase and higher ratings than the tactile and 3-D audio displays, respectively, in the final phase. This corresponds to the best of both worlds because of the nonsignificant difference with the best included display at each threat phase. Second, as indicated above, in terms of redundancy effects there is no indication of increased initial alerting and reduced RT (no gestalt) with the trimodal display. Third, the trimodal display did not lead to increased mental workload or interference, as indicated by the subjective ratings of mental workload and effort of driving and performance on the PCR secondary task.

EXPERIMENT 2

Experiment 1 demonstrated important beneficial effects of the trimodal display on overall perception and performance in the dynamic threat scenario. The main changes for Experiment 2 were that the unimodal conditions of visual and tactile threat direction cueing were replaced by bimodal conditions, the unimodal 3-D audio condition was excluded, and the HDD was replaced by a HUD. First, even though overall performance with the trimodal display was superior in Experiment 1, there were in terms of gestalt effects primarily no enhanced alerting or reduced RT with the trimodal display. An explanation may be that the trimodal display was not optimized for these effects in that the HDD cue was presented at another location than the 3-D audio and tactile cues. That is, further enhancement effects in terms of gestalt of multisensory presentation may not have appeared because of using visual stimuli deviating from the general spatial synchrony of the 3-D audio and tactile stimuli (Santangelo & Spence, 2007; Spence & Driver, 2004; Spence & Ho, 2008). Furthermore, the HDD presentation consisted of an allocentric view in contrast to the egocentric auditory and tactile presentations. Thus, the change from an allocentric HDD to an egocentric HUD was made in an effort to improve the trimodal spatial synchrony and equalize the information coding of the displays. (Note also that although the unimodal threat direction cueing with the HDD and the tactile belt each paired with the nondirectional auditory alert may have caused positive effects, it may also have caused some negative effects because of spatial asynchrony.) Second, although the combination of tactile belt and HDD can be considered necessary and sufficient for the improvement of the threat cueing over the unimodal displays in Experiment 1, the solution for implementation in the combat vehicle would be the trimodal display including 3-D audio. This is so because an auditory alert will not be excluded for prioritized threats in the combat vehicle. Using 3-D audio instead of nondirectional auditory alert has several potential advantages; it not only makes it possible to present 3-D localized sound but also makes it

more probable for beneficial multisensory effects of spatial synchrony with information employing other sense modalities (e.g., Ho, Santangelo, & Spence, 2009). Thus, all the experimental conditions in Experiment 2 involved display combinations with 3-D audio. Along these lines, the bimodal conditions in Experiment 2 consisted of HUD with 3-D audio and tactile belt with 3-D audio, and the trimodal display combined HUD, 3-D audio, and tactile belt. Third, another explanation for no gestalt effect of reduced RT with trimodal display in Experiment 1 may be that the perceptual or mental workload was too low (e.g., Santangelo & Spence, 2007). The secondary task was thus modified with the purpose to increase the perceptual or mental workload induced in the participants. Again, with the same threat cueing scenario as in Experiment 1, the main hypotheses were that the trimodal display leads to overall best performance and overall best perception of direction to threat.

Method

Participants. A total of 12 naïve participants, 5 females and 7 males with normal sight and hearing, participated. Their mean age was 25.8 years (21–34 years).

Apparatus and stimuli. The apparatus was the same as in Experiment 1, except that a HUD was simulated on the 37-in. plasma screen and no HDD was used for visual threat cueing.

The experiment had a 3 (display) $\times$ 3 (sector) within-subjects design. It included three displays for threat direction cueing composing HUD with 3-D audio, tactile with 3-D audio, and HUD, 3-D audio, and tactile belt combined into a trimodal display (Table 2). The HUD consisted of a rectangle, drawn in outline, with a pulsing arrow on either side indicating direction to threat and with decreased pulse speed of the arrow indicating reduced angular distance to threat. When threat was within the view of the windshield in the out-of-the-window view, the arrow moved inside the rectangle that dynamically centered on the threat. The HUD is presented in Figure 5.

The touch screen used for response of radio calls was reconfigured with 12 buttons in two

TABLE 2: The Three Display Conditions With Type of Threat Cueing and Modality for Radio Calls

Display	Threat Direction Cueing and Sound Alert	Radio Calls
HUD–3-D audio	HUD + 3-D audio	3-D audio
Tactile–3-D audio	Tactile belt + 3-D audio	3-D audio
Trimodal	HUD + tactile belt + 3-D audio	3-D audio

Note. HUD = head-up display.

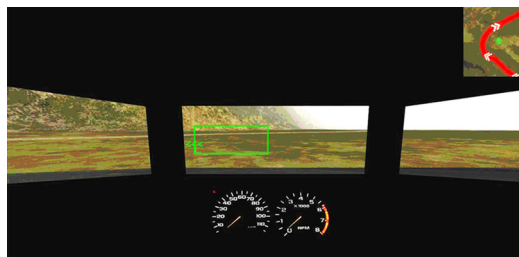


Figure 5. The simulated combat vehicle and surrounding terrain as viewed from driver position. The head-up display symbology in the center consisting of a rectangle and arrows indicates a threat to the left of the vehicle.

rows. The upper row presented 4 buttons, each in a specific color and the name of the color printed on each button. The lower row presented 8 numbered buttons (1–8). The radio calls consisted of four sequentially and partially overlapping voices. Each voice conveyed information on name, color, and number, and the task was to identify the color and number for the voice initiated with the name Caesar. The secondary task response was to push the corresponding buttons on the touch screen. Analogous to Experiment 1, the radio calls were presented simultaneously with threat cueing onsets, and the secondary task response was performed after completion of the main task. For all three display configurations, the radio calls were presented in 3-D audio with the voices positioned horizontally at -45° , -22° , $+22^\circ$, and $+45^\circ$ from the center. The positions of voices were randomized for each presentation. The reason for the modification was the high accuracy on the secondary task in Experiment 1, which seemed to impose little mental workload. The modification ensured that it provided extra load on the employed sensory (auditory, visual, tactile, and motor) systems.

The threat cueing onset in three sectors was in principal identical with Experiment 1, but with the differences that each sector was reduced to $\pm 40^\circ$ and that there was a gap of $\pm 10^\circ$ at the center of the front sector. This central gap made it necessary to align the vehicle with all threats, before pushing the trigger button. Each participant was exposed to three blocks of trials, one block for each display configuration. The block order was balanced across participants.

In all other respects, the threat scenario was identical to Experiment 1. A question about to which degree the participants used the 3-D audio sound for threat localization for each display was included in the final questionnaire. Otherwise, the questionnaires were identical to the ones used in Experiment 1.

Procedure. The procedure was the same as in Experiment 1, except that the training–familiarization session was extended to three threat presentations for each display. Total time for each experimental session was approximately 75 min.

Results

In the case of violation of the sphericity assumption in the ANOVAs, the Greenhouse-Geisser-corrected p value is given (indicated by ^G). All post hoc tests were performed with Tukey’s HSD.

Performance data. Performance data were analyzed with ANOVA for a 3×3 repeated measures factorial design (3 displays $\times$ 3 sectors). A few participants showed a deviant performance for some measures on their first block of trials. No such deviations were found in the following blocks, which indicate that they would likely have needed more training. Therefore, measurements in the first block deviating more than two standard deviations from the mean of respective condition were replaced by the group mean for that condition. In the total of 108 data points

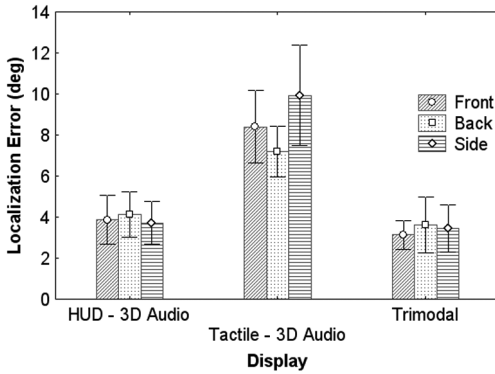


Figure 6. Mean localization error with each type of display configuration. Error bars = ± 1 standard error; HUD = head-up display.

(12 participants $\times$ 3 displays $\times$ 3 sectors) for each dependent variable, the following corrections were made: RT = 3 corrections, LE = 6 corrections, and PCR = 4 corrections. Before analysis with ANOVA, the PCR data were, because of proportions near .80, transformed with the 2arcsine(x) formula to remedy violations against the assumption of normal distribution (Kirk, 1995).

Localization error. The ANOVA showed a main effect of display, $F(2, 22) = 46.36, p < .001$, and a significant interaction effect of display by sector, $F(4, 44) = 5.29, p < .01$ (Figure 6). There was no significant main effect of sector. Post hoc testing showed that the main effect of display was the result of a significantly larger LE for the tactile-3-D audio display, both compared with the HUD-3-D audio and the trimodal displays ($ps < .001$). Post hoc testing also showed that the interaction effect was the result of a small but significantly larger LE with the tactile-3-D audio display in the side sector compared with the back sector ($p < .001$).

Response time. The ANOVA showed main effects of display, $F(2, 22) = 8.59, p < .01^G$, and sector, $F(2, 22) = 33.19, p < .001^G$, but no significant interaction effect. Post hoc testing showed that the tactile-3-D audio display ($M = 1987, SE = \pm 98$ ms) had a significantly longer RT than both the HUD-3-D audio (1682, ± 42 ms) and trimodal display (1712, ± 71 ms; $ps < .01$). Post hoc testing also showed that the RT for the front sector (1954, ± 57 ms) was significantly

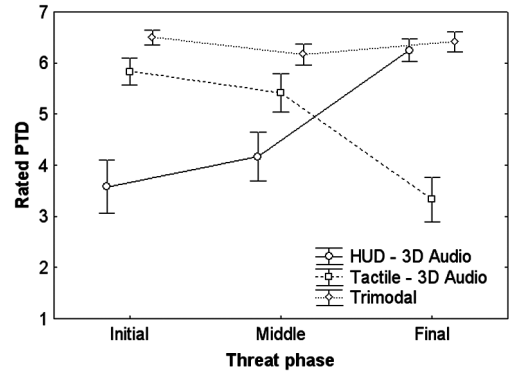


Figure 7. Perception of threat direction (PTD) for the three display types at threat cue onset (initial), maneuvering toward threat (middle), and aligning vehicle with threat (final). Error bars = ± 1 standard error; HUD = head-up display.

longer than for the back (1718, ± 60 ms) and side (1709, ± 63 ms) sectors ($ps < .001$).

Proportion correct radio calls. The ANOVA on 2arcsin(x)-transformed values showed no significant effects of PCR. The overall mean PCR was 0.79 (untransformed).

Subjective ratings. Perception of threat direction was analyzed with ANOVA for a 3×3 repeated measures factorial design (3 displays $\times$ 3 threat phases). Mental workload, effort of driving, and use of the 3-D audio were analyzed with ANOVA for a one-way repeated measures design (3 displays).

Perception of threat direction. The ANOVA showed a main effect of display, $F(2, 22) = 17.88, p < .001^G$, and a significant interaction effect of display by threat phase, $F(4, 44) = 14.58, p < .001^G$. There was no significant main effect of threat phase (Figure 7).

Post hoc testing showed that the main effect of display was the result of significantly higher ratings of the trimodal display compared to both the HUD-3-D audio and the tactile-3-D audio displays ($ps < .001$). Post hoc testing showed the following significant differences of the PTD ratings of the display configurations within each threat phase: initial—both trimodal and tactile-3-D audio higher rated than HUD-3-D audio ($ps < .01$); middle—trimodal higher rated than HUD-3-D audio ($p < .01$); and final—both

trimodal and HUD-3-D audio higher rated than tactile-3-D audio ($ps < .001$). Post hoc testing also showed that the PTD ratings of the HUD-3-D audio display were significantly lower in the initial and middle phases compared with the final phase ($p < .001$ and $p < .01$, respectively) and that the PTD ratings of the tactile-3-D audio display were higher in the initial and the middle phases compared with the final phase ($ps < .001$).

Mental workload. The ANOVA showed no main effect of display. The ratings of mental workload were moderate for all three displays with an overall mean of 3.9 on the 7-point rating scale (1 = *very low*, 7 = *very high*).

Effort of driving. The ANOVA showed no main effect of display. The ratings of effort of driving were low for all three displays with an overall mean of 2.1 on the 7-point rating scale (1 = *very low*, 7 = *very high*).

Use of 3-D audio. The ANOVA showed no effect of display, and the overall mean of rated use of the 3-D sound was low, at 1.9 on the 7-point rating scale (1 = *did not use the sound at all*, 7 = *did use the sound very much*).

Conclusions

First, the main hypothesis that overall performance would be best with the trimodal display was not supported because the trimodal display was equal to the best bimodal display, the HUD-3-D audio, for both of the two performance measures. In other words, both LE and RT with the trimodal display can be explained by the contribution of the HUD-3-D audio. In terms of redundancy effects, this corresponds to the best included bimodal display—the best of both worlds for each performance measure. Second, the main hypothesis that PTD would overall be best with the trimodal display was supported. This can be explained by the tactile-3-D audio display in the initial and middle threat phases and by the HUD-3-D audio in the final threat phase—the best of both worlds for each phase. However, since the main effect of display was the result of significantly higher PTD ratings of the trimodal display compared with both the HUD-3-D audio and the tactile-3-D audio, there was a gestalt effect of threat perception of using the trimodal display. Third,

as indicated earlier, in terms of gestalt the trimodal display did not entail any effects of increased initial alerting or reduced RT because the best bimodal display for RT (i.e., HUD-3-D audio) and initial PTD (i.e., tactile-3-D audio), respectively, can explain the trimodal effects at the initial threat phase (i.e., no gestalt effects at the initial threat phase). It is, however, important to note that in terms of overall performance and perception, the trimodal display was superior in the initial phase by providing shorter RT than the tactile display and better initial threat perception than the HUD. Fourth, as indicated by rated mental workload and effort of driving and performance in the PCR secondary task, the trimodal display did not lead to interference or increased mental workload. Finally, the use of the 3-D audio at threat cueing was considered low.

GENERAL DISCUSSION

Overall, Experiments 1 and 2 reveal important benefits of using trimodal display configurations for cueing of direction to threat in dynamically framed threat scenario. Fundamental is that different phases of the used threat scenario put different requirements on the operator depending on whether it was the initial phase of first perceiving direction to threat, the middle phase of maneuvering toward threat, or the final phase of precision maneuvering. PTD was overall best with the trimodal cueing in both experiments, revealing that the included display best suited for each phase of threat direction cueing was primarily utilized (i.e., initially tactile belt and finally HDD or HUD). Overall performance was also best with the trimodal display in Experiment 1, which was explained by the contribution of the tactile display to RT and the HDD to LE. The shorter RT with the trimodal display compared to the HDD in Experiment 1 illustrates the advantage of increased alerting by including redundant tactile information when visual information is presented on a HDD, which is not constantly attended. In Experiment 2, the bimodal HUD-3-D audio display was on par with the trimodal display for both performance measures of the main task, which indicates that the HUD-3-D audio is sufficient for the improved overall performance (as measured here). However, if

TABLE 3: Summary of Group Means of the Measures for the Displays in Each Experiment

Exp.	Display	Measure							
		LE	RT	PCR	PTD Initial	PTD Middle	PTD Final	MW	ED
1	HDD	2.8*	2,759	.85	4.2	5.2	6.4*	3.8	2.3
	3-D audio	6.8	2,810	.81	3.6	4.3	3.9	4.4	2.3
	Tactile	6.3	2,175*	.82	6.7*	6.2	4.8	4.0	2.1
	Trimodal	3.0*	2,253*	.85	6.7*	6.7*	6.7*	3.8	2.3
2	HUD – 3-D audio	3.9*	1,682*	.85	3.6	4.2	6.3*	4.1	2.0
	Tactile – 3-D audio	8.5	1,987	.74	5.8*	5.4	3.3	3.6	2.3
	Trimodal	3.4*	1,712*	.79	6.5*	6.2*	6.4*	4.1	1.8

Note. LE = localization error (degrees); RT = response time (ms); PCR = proportion of correct radio calls; PTD = perception of threat direction; MW = mental workload; ED = effort of driving (subjective ratings, scale = 1–7); HDD = head-down display; HUD = head-up display.
*Indicates significantly best display configuration(s) for each measure within each experiment.

visual distraction had occurred, and thus continuous attention to the HUD was not possible, redundant tactile information may also have contributed to shorter RT with the trimodal display in this experiment. Nevertheless, because PTD was rated higher with the trimodal display than with the HUD–3-D audio in both the initial and middle phases, the trimodal display can be considered overall advantageous also in this experiment. Taken together, the trimodal display in both experiments overall showed the most advantages, as indicated by Table 3.

Although the actual performance certainly is of great importance, high quality of the perceived threat direction may be a precursor to improved performance in more extreme conditions with high physical, perceptual, or cognitive loads. Thus, improved PTD may have a decisive impact on comprehension and performance in threatening situations with more severe conditions of driving a combat vehicle. An example is the quality of perception of threat cueing as related to performance improvement with the addition of tactile cueing to visual cueing in aerial combat with high acceleration during threat intercept in van Erp et al. (2007; also see Eriksson et al., 2006). Furthermore, it is likely that a tactile display would increase performance and ease mental workload in situations with high mental workload and when vision is more occupied (e.g., Gilson et al.,

2007; Ho, Tan, & Spence, 2005; Redden, 2006; van Erp, Veltman, van Veen, & Oving, 2003). The real benefits of tactile direction cueing may thus be demonstrated “given conditions of high cognitive workload, multiple visual demands, and/or degraded or limited visual channels” (Redden, 2006, p. 1699). Because the contribution of the 3-D audio to the trimodal display seems to have been weak in both experiments, it may be sufficient to combine visual and tactile displays for improving display effectiveness. Still, trimodal displays employing vision, audition, and touch may show advantages over unimodal or bimodal displays in presenting prioritized information. That is, it cannot be ruled out that auditory displays add value to visual and tactile displays in certain conditions, and especially when visual or tactile information is not accessible (Ho et al., 2009; Santangelo & Spence, 2007; Self, van Erp, Eriksson, & Elliott, 2008; Veltman et al., 2004). For prioritized threat cueing in a combat vehicle, it may thus be best to include visual and tactile displays together with an auditory display, not the least because one or two of the displays may not be readily available or perceived at a given crucial moment. Cross-modal links also have been demonstrated for vision, audition, and touch, and temporally and spatially synchronized multimodal events maximally activate multisensory neurons in the cortex (Spence & Driver, 2004),

which may not even require a perfect spatial synchronization (e.g., Spence & Ho, 2008). Thus, a trimodal display employing visual, auditory, and tactile cueing for redundant information may prove to be overall superior in promoting safer and more reliable peak performance in certain time critical tasks.

Although the trimodal redundancy effects characterized as the best of both worlds were obvious in the experiments, no gestalt appeared, except concerning PTD in Experiment 2, where PTD with the trimodal display was rated overall higher than both the HUD-3-D audio and the tactile-3-D audio displays. However, there were no other gestalt effects, and there may be at least three possible explanations for this: too low perceptual or mental workload, all conditions being multimodal, and multimodal spatial synchrony that was not good enough. First, it is plausible that we did not properly manage to induce a high enough level of mental workload, which may explain the lack of manifest gestalt effects of the trimodal displays. Second, almost all of the displays in the experiments were in fact multimodal. In Experiment 1 the unimodal directional cueing was made in combination with nondirectional sound, except for the 3-D audio condition, and in Experiment 2 the trimodal display was compared with bimodal displays. This may thus explain that no further gestalt effects appeared. Third, an important aspect is that of spatial synchrony of the multisensory stimuli, and we may not have managed to achieve this properly. Thus, gestalt effects may not have shown up because of some spatial asynchrony of the sensory stimuli that did not engage multimodal cortex centers to a sufficient degree.

Although the 3-D audio display provided precision comparable to the tactile display in Experiment 1, it suffered from front-back confusion, and the subjective ratings indicate that it was relatively difficult to perceive threat direction with the 3-D audio display. The RT was longer than with the tactile display, which probably was connected to the difficulty of perceiving initial threat direction, reflected by the subjective PTD ratings. Although this may have been an effect of interference from the simultaneous information coding of the radio calls,

performance of the PCR task and ratings of the mental workload were comparable to those for the other displays. Furthermore, mean perceived accuracy of threat position with 3-D audio, with no secondary task, in Carlander and Eriksson (2006) was very similar to the ratings of perceived threat direction in Experiment 1 of the present study. Improvements of the 3-D audio may involve alleviating the front-back confusion problem by displacing presentations from the 180° position straight behind or by using another frequency for presentations in the back sector. More important, however, is that the 3-D audio can be improved by using another better-adapted alarm sound and individually tailored HRTFs.

It is not surprising that both the HDD and the HUD-3-D audio provided the best precision in the final threat phase, besides the trimodal displays. However, perceived threat direction was rated low for both the HDD and the HUD-3-D audio at the initial threat phase with threat cueing onset. For the HDD this was likely mainly the result of the operator's need of changing visual focus. For the HUD, with less operator need of changing focus, this could reasonably been alleviated by a display design with explicit presentation of direction to threat. The HUD-3-D audio nevertheless showed RT not significantly different with the RT for the trimodal display. Since RT with unimodal 3-D audio in Experiment 1 was relatively long and on par with HDD, improved RT with the HUD compared to HDD was likely mainly the result of the change from HDD to HUD and not the inclusion of 3-D audio with the HUD. The subjective ratings in Experiment 2 also indicate little use of 3-D audio for threat cueing. It is however unfortunate that the experimental design did not include the bimodal condition of HUD combined with tactile belt in Experiment 2. It thus cannot be ruled out that HUD combined with tactile belt would have been as beneficial as the trimodal display in terms of performance and perceived direction to threat. Nevertheless, as mentioned earlier, auditory displays may of course add value to visual and tactile displays.

The tactile display provided a distinct indication of threat position at initial and middle threat

phases, which indicates the intuitiveness of a tactile display covering the torso and enabling dynamic direction information around the display wearer. However, the precision of the tactile display was lower than for both the visual and trimodal displays, but this is more or less normal considering the design of the tactile belt. On the other hand, with each tactor covering 30° of the horizontal dimension, the precision with the tactile display was interestingly high, in the range of 6° to 8°. The reason seems to be that the dynamics of moving the vehicle, and thus moving the threat indication from tactor to tactor, make it possible to integrate the time factor. As applied to the task in the present experiments, the precision of the tactile display can be enhanced by adding more tactors close to the central tactor on the frontal torso. For example, van Erp (2005b) showed that the spatial acuity was about 2–3 cm uniformly over the torso, except for the body midline, where it was found to be about 1 cm for horizontal arrays. Important also is the direction acuity, or the mapping of body coordinates to external directions, and van Erp (2005a) suggests the use of a torso-related transfer function for precision tasks because of measured biases up to 10° in the perceptual transforming of tactile indications to external directions. Nevertheless, our results mainly corroborate the results of Carlander and Eriksson (2006), which also showed that despite using a tactile display resolution of 30° it is possible to get errors of localization below 10° in the employed dynamic task.

An objection to multimodal displays is that simultaneous presentation to different senses might impose higher mental workload and thereby reduce mental spare capacity for simultaneous handling of other tasks. However, as compared with uni- and bimodal displays, the results do not indicate increased mental workload of using trimodal displays, nor interference with the auditory loading secondary task. The similar results for all displays on the secondary task indicate that the participants followed the instructions for all displays, which also implies validity of the main task. In this sense, the secondary task provided a manipulation check. Furthermore, if the complete dynamic task of threat handling is considered with the view that

perception and performance of the whole task are of greater importance than are perception and performance of its parts, then the overall better perception and performance with the trimodal display may be expressed in terms of a general gestalt-related effect. Above all, the results demonstrate that multimodal displays with sensory redundant information can overcome display-specific limitations and provide enhanced perception and performance in a dynamically framed threat scenario.

ACKNOWLEDGMENTS

The experiments were performed within research projects at the Swedish Defence Research Agency. The Swedish National Road and Transport Research Institute and the Swedish Defence Research Agency funded work on this article. Experiment 1 was presented at Human-Computer Interaction International 2007 (Carlander, Eriksson, & Oskarsson, 2007), and Experiment 2 was presented at the 2008 Human Factors and Ergonomics Society annual meeting (Oskarsson, Eriksson, Lif, Lindahl, & Hedström, 2008). We gratefully acknowledge the reviewers and the editor for invaluable comments and suggestions for improvements. Staffan Nählinder, Swedish Defence Research Agency, is thanked for valuable advice.

KEY POINTS

- The trimodal combinations of visual, auditory, and tactile displays provided overall best perception of threat direction, and the overall performance was paralleled only by the HUD–3-D audio in Experiment 2. This indicates that combining displays into a multimodal display can overcome display-specific limitations and enhance operator perception and performance.
- The HDD and HUD provided high precision but lower perception of threat direction in the initial phase of the dynamic threat scenario.
- The tactile display provided better perception of threat position in the initial threat phase compared to the visual displays, which indicates the intuitiveness of tactile information distributed over the torso and covering directions around the wearer of the display.
- The 3-D audio was the display that seemed to contribute least to the trimodal displays.

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- Date received: August 12, 2010*
Date accepted: September 2, 2011